**VELLore INSTITUTE OF TECHNOLOGY**

School of Electronics Engineering (SENSE)

# Measurement of Heart Rate (BPM) and Oxygen Saturation ($SpO_2$) Using Arduino (MAX30100 Sensor)

Signals and Systems Project Report

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## 1. Abstract

This project aims to design and implement a simple biomedical monitoring system that measures a person's heart rate (BPM)
and oxygen saturation level ($SpO_2$) using the MAX30100 pulse oximeter sensor interfaced with an Arduino Uno microcontroller.
The system provides real-time monitoring and displays the readings on an LCD screen. It serves as an efficient low-cost
health monitoring prototype suitable for wearable and remote healthcare systems.

## 2. Introduction

Heart rate and blood oxygen level are two critical physiological parameters for assessing a person's health condition.
This project utilizes the MAX30100 sensor module that integrates two LEDs (red and infrared) and a photodetector to measure
pulse and oxygen levels. The Arduino Uno processes the sensor data and displays the results on a 16x2 LCD screen.
This system demonstrates signal acquisition, filtering, and display principles studied in the Signals and Systems course.

## 3. Components Used

1. Arduino Uno
2. MAX30100 Pulse Oximeter Sensor
3. 16x2 LCD Display
4. Breadboard and Jumper Wires
5. USB Cable for Power and Data

## 4. Circuit Connections

- MAX30100 VCC → 3.3V on Arduino
- MAX30100 GND → GND
- MAX30100 SDA → A4 (SDA)
- MAX30100 SCL → A5 (SCL)
- LCD VCC → 5V
- LCD GND → GND

- LCD SDA → A4
- LCD SCL → A5

## 5. Working Principle

The MAX30100 sensor emits red and infrared light through a fingertip. The photodetector measures the reflected intensity,
which varies with the pulsatile flow of blood. Using signal processing algorithms, the Arduino extracts the heart rate and
$SpO_2$ values. The BPM and oxygen saturation values are updated on the LCD screen in real-time.

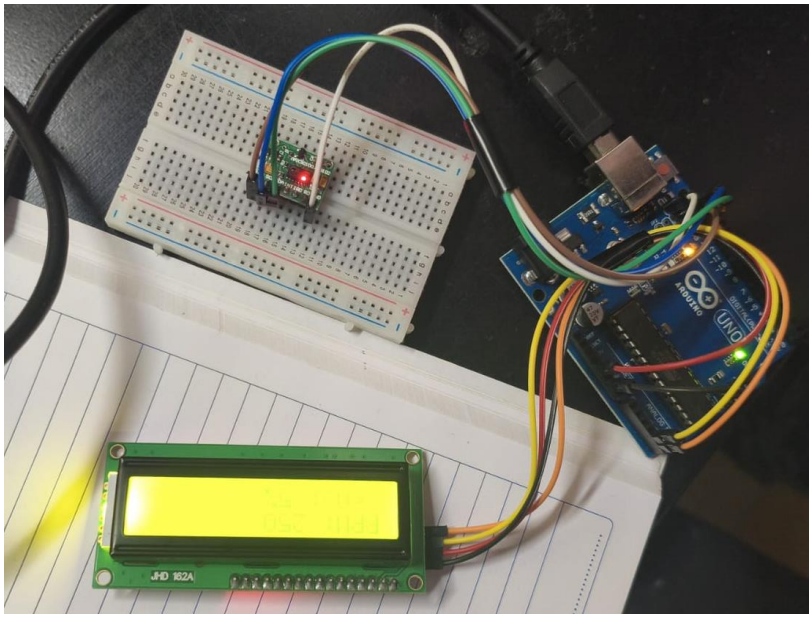
## 6. Arduino Code

```
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
PulseOximeter pox;

uint32_t tsLastReport = 0;

void onBeatDetected() {
  lcd.setCursor(0, 0);
  lcd.print("Beat!");
}

void setup() {
  Serial.begin(115200);
  lcd.init();
  lcd.backlight();
  if (!pox.begin()) {
    lcd.print("Sensor Error");
    for (;;);
  }
  pox.setOnBeatDetectedCallback(onBeatDetected);
}
```

```
void loop() {
  pox.update();
  if (millis() - tsLastReport > 1000) {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("BPM: ");
    lcd.print(pox.getHeartRate());
    lcd.setCursor(0, 1);
    lcd.print("SpO2: ");
    lcd.print(pox.getSpO2());
    lcd.print("%");
    tsLastReport = millis();
  }
}
```

## 7. Results and Observations

The prototype successfully displayed heart rate and oxygen saturation levels on the LCD screen.
Sample readings observed were approximately 75 BPM and 97% $SpO_2$ under testing conditions.
The results demonstrate accurate detection and real-time display capabilities.



## 8. Conclusion

The designed system effectively measures BPM and $SpO_2$ levels using the MAX30100 sensor and Arduino.
This project highlights the practical application of signal processing concepts in biomedical instrumentation.
Future enhancements could include wireless data transmission or integration with mobile health apps.

## 9. References

1. MAX30100 Datasheet – Maxim Integrated
2. Arduino Documentation – www.arduino.cc
3. GitHub – MAX30100 Pulse Oximeter Library